SANKALPA RAYMONDAL

Academic CV

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IDENTITY

I am a highly motivated and determined individual with a strong passion for robotics and its potential to serve society. Driven by a love of learning and a commitment to excellence, I aspire to contribute to advancements in the field.

EDUCATION

Nanyang Technological University, Singapore, **Aug 2020 – May 2024**
Bachelor of Engineering (Mechanical Engineering)

- Specialization in Mechatronics and Robotics
- Honours (Highest Distinction) | CGPA 4.63/5
- Dean's List for Academic Year 2021/2022 and 2022/2023
- Relevant courses: Robotics, Microprocessors Systems, Measurements and Sensing Systems, Systems Thinking and Systems Engineering, Mechatronics System Interfacing, Control Theory, Electrical Circuits and Electronic Devices, Introduction to Computational Thinking, Mathematical Methods in Engineering, Venturing into Entrepreneurship, Introduction to Biomolecular Engineering

Delhi Public School – Modern Indian School, Qatar, **Apr 2016 – May 2020**
Senior School Certificate Examination, 2020

- Achieved 95% marks overall
- Subjects: English core, Mathematics, Physics, Chemistry, Computer science

WORK EXPERIENCE

Micron Semiconductor Asia Operations Pte Ltd **Sep 2024 – Present**

Shift Equipment Engineer
Manage the operations of Scanner Photolithography equipment (ASML and Canon) in a 24/7 fabrication facility ensuring safety, quality and peak performance

Troubleshoot and analyze equipment issues to enable quick recovery; escalate recurring problems to tool owners with structured problem statements and background information

Collaborate with cross-functional stakeholders - each with distinct priorities - to maximize line efficiency through clear communication and balanced, data-driven decision-making

Supervise 3-5 equipment technicians and allocate manpower effectively; present shift summaries of tool status and major issues in daily handover meetings

Lionsbot International Pte Ltd **Jan 2023 – Jun 2023**

Validation Intern (Quality Assurance)
Worked as a validation intern for an autonomous cleaning robot company.

Rapid Prototyping:

Measured parts, determined suitable tolerances, designed parts on Solidworks and then used 3D-printing, laser-cutting or machining (drilling, chamfering) to rapidly assemble reliable prototypes.

Implemented relay modules, various sensors and actuators for various testing jigs like Hose Clamp Durability and Vacuum Motor Performance.

Test environment creation:

Planned and wrote test specifications for performance requirements, managed tests from setup to execution, documented the procedures and outcomes, maintained validation schedules and liaised with key-stakeholders on test equipment setup and availability.

Developed Python scripts to read data from sensors connected to Arduino and generated graphs from recorded data.

Repurposed existing setup of a Wheel-Testing jig after significant adjustments for testing the fatigue on Cleaning-Tool Wheels.

Validation:

Validated general performance of robot by running it for long periods of time, and validated software changes by physically creating potential scenarios to test the robot.

Reported a wide variety of problems to relevant departments ranging from hardware issues like abnormal spray pattern to software problems in Navigation, Perception and Localization with detailed description.

Introduced to ROS in a Linux environment and used simple commands to visualize the perception of the robot after connecting to the robot through TeamViewer.

ACADEMIC PROJECT WORK

Final Year Project – Physics simulation of human and exoskeleton,
Jul 2023 – May 2024

Developed human-in-the-loop physics simulation of a single sided exoskeleton to assist patients with mobility related disabilities in walking building upon existing work done by Rehabilitation Research Institute of Singapore (RRIS).

Conducted extensive literature review to understand existing research and simulation methods.

Converted Solidworks model of exoskeleton to URDF format suitable for simulation.

Explored and utilized the features of MuJoCo, the employed physics engine, to enhance simulation accuracy.

Optimized exoskeleton-fitting to ensure stability and realistic human interaction.

Gained proficiency in Linux environment for coding and Github for code management.

Systems Thinking and Engineering – Enhanced Netflix services proposal, Aug 2023 – Dec 2023

Performed a comprehensive systems analysis to identify and propose improvements for Netflix's existing services.

Conducted stakeholder analysis, and defined the acceptance criteria, functional/operational requirements and impact.

Utilized context diagrams, power-interest grid, Pugh matrix, use case diagrams, risk matrix for assisting in analysis.

Robotics – Tic-tac-toe playing robotic arm, Aug 2023 – Dec 2023

Contributed to the development of a robotic arm with computer vision capable to play tic-tac-toe with human.

Evaluated the suitability of MoveIt2 with ROS2 for the project by exploring tutorials. Aided in preparing project report.

Microprocessors – Class assessment, Aug 2022 – Dec 2022

Prepared breadboard for assessment by soldering several parts onto it.

Programmed ARM-based microcontroller to use its GPIO and PWM features for switching on LEDs, controlling buzzers and reading values from DIP switches.

Engineering Innovation and Design (EID) – Thermate, Jan 2022 – May 2022

Initiated ideation, designing and testing of a cup heater that keeps hot beverages warm for a longer time.

Acquired necessary electrical components from vendors and ensured price and quality.

Yielded simulations on liquid's temperature with and without cup heater using Simulink in MATLAB. Documented product development for business plan and presented it to judges.

Data Science and Artificial Intelligence – Predicting earthquake damages, Jan 2022 – May 2022

Used logistic regression, random forest and naïve bayes classifiers from scikit-learn to find the best earthquake damage classification using data from "Richter's Predictor: Modelling Earthquake Damage", Driven Data and presented the findings.

Optimized parameters like number of iterations to find the highest cross validation score of the classifiers. Directed the use of a pipeline from scikit-learn that combines One Hot Encoding, machine learning optimization and cross validation for simplification.

Machine Element Design – Sugarcane crushing machine design, Jan 2022 – May 2022

Designed a Sugarcane Crushing Machine by following design principles and doing necessary calculations for the Belt drive, Chain drive, Gear train, Shafts, Bearings and Retaining rings comprising it.

Implemented Solidworks to make a 3-D model of Belt drive and Chain drive.

URECA – Glove controller for insect-computer hybrid robot, Aug 2021 – Jun 2022

Designed a glove controller using Arduino and piezo-film sensors to control the movement of an insect-computer hybrid robot.

Created a research poster and a report for the project after learning about the required procedures for research.

Mechatronics System Design - Medicine vending machine, Aug 2021 – Dec 2021

Developed a prototype of a medicine dispensing mechanism by programming the integration of actuators and sensors using Arduino microcontroller.

Conducted extensive testing, adjusted parameters, and fixed bugs to ensure smooth functioning.

Achieved silver position along with other teams in the course.

Computer Sc. project –Railway reservation system in C++, Apr 2019 – Feb 2020

Planned the implementation of a Railway Reservation System and created a basic structure of the program.

Applied concepts of Object-Oriented Programming, Classes, Inheritance, Data File Handling, etc.

SKILLS

Technical Skills:

- C++,
- Python,
- Fusion 360,
- Solidworks,
- 3D printing,
- Machining,
- Arduino,
- MATLAB,
- Simulink,
- Microsoft Office,
- scikit-learn,
- Communication skills,
- Leadership skills.

Languages: English (Bilingual Proficiency), Bengali (Native), Hindi (Working Proficiency), French (Elementary Proficiency), German (Elementary Proficiency)

CERTIFICATION

- University of Toronto – Introduction to Self-Driving Cars (Coursera), Aug 2023
- UC San Diego - Algorithmic Toolbox (Coursera), Mar 2023
- LinkedIn Learn - Python Data Structures and Algorithms, Sept 2022
- University of Michigan - Successful Negotiation: Essential Strategies and Skills (Coursera), Jan 2022
- Yale University - Financial Markets with Honors (Coursera), Jul 2022
- Stanford University - Machine Learning (Coursera), Jul 2021

CO-CURRICULAR ACTIVITIES

Honorary General Secretary for NTU Cricket Club, Jun 2023 – May 2024

Founded club with fellow students, planned and organized events like Orientation Week, Cricket Recreational Matches, and generated posters for publicity. Liaised with university's administrative authorities, pivoted direction of club as required and delivered required administrative work. Provided clear ideas on how to implement events and performed the required work to accomplish it.

Chess Captain for Hall 11, **Aug 2021 – May 2022**
Conducted regular chess training sessions. Represented residential hall in university for inter-hall chess games.

Peer Tutor for Mechanical and Aerospace Engineering, **Aug 2021 – May 2022**
Taught Mathematics I and Mathematics II to juniors on a weekly basis as a volunteer. Cleared doubts and provided advice in doing better in course.

HOBBIES AND INTEREST

Reading, Duolingo (learning German), Cricket, Chess, Table Tennis, Playing Harmonica, Movies, Music.

REFERENCES

Assoc Prof Ang Wei Tech – Nanyang Technological University Singapore, wtang@ntu.edu.sg

Prof Hirotaka Sato – Nanyang Technological University Singapore, hirosato@ntu.edu.sg

Assoc Prof Sameer Alam – Nanyang Technological University Singapore, sameeralam@ntu.edu.sg